

HARTZ MOUNTAINS, AustraliaWMO: **949770**Lat: **43.201S**Lon: **146.768E**Elev: **2726**StdP: **13.30**Time Zone: **10.00 (AUS)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.9	29.9	24.5	20.6	41.5	27.4	23.6	34.7	46.4	38.7	40.5	37.0	20.7	250	0.892

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.9	75.7	59.8	71.0	58.0	67.3	56.6	61.5	71.2	59.6	68.5	57.8	65.5	13.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.1	79.8	64.3	56.1	74.2	62.6	54.3	69.3	60.7	28.8	71.1	27.4	68.7	26.1	65.9	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
40.5	35.7	32.2	DB	27.0	85.3	1.2	4.1	26.2	88.3	25.5	90.7	24.8	93.1	24.0	96.1	
			WB	26.9	65.2	1.2	1.3	26.0	66.2	25.4	66.9	24.7	67.7	23.9	68.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	52.7	52.4	49.6	45.5	42.0	39.5	37.9	38.2	40.1	43.0	46.7	48.7
	DBStd	7.96	7.48	7.50	7.19	5.79	4.73	3.93	3.69	4.63	5.15	6.48	7.51	7.39
	HDD50	2477	53	55	96	161	252	316	374	366	299	237	153	115
	HDD65	7452	390	359	480	586	713	766	839	831	746	682	552	508
	CDD50	525	136	122	85	25	5	0	0	1	3	20	53	75
	CDD65	23	8	6	4	0	0	0	0	0	0	0	2	3
	CDH74	195	85	47	22	0	0	0	0	0	0	1	13	27
	CDH80	36	18	7	5	0	0	0	0	0	0	0	1	5
	WSAvg	16.9	15.8	14.3	15.3	16.4	18.3	17.9	17.8	18.0	18.7	17.4	16.5	16.1
	PrecAvg	39.3	2.5	2.2	2.5	2.8	3.3	3.6	3.9	4.8	4.0	3.8	3.1	2.9
(15) Precipitation	PrecMax	50.7	5.9	6.6	5.0	7.6	9.1	7.2	7.8	8.7	7.8	7.8	5.5	5.8
	PrecMin	31.6	0.4	0.7	0.8	0.9	0.7	1.3	1.6	1.4	1.3	1.1	0.6	0.6
	PrecStd	5.0	1.4	1.5	1.1	1.4	2.0	1.6	1.7	1.9	1.8	1.5	1.2	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.5	80.3	76.5	65.2	57.5	52.4	50.0	54.7	59.6	69.2	75.5	77.8
		MCWB	61.3	62.4	59.8	54.2	49.4	47.9	44.5	47.5	50.0	53.7	57.2	59.4
	2%	DB	75.6	73.4	69.0	60.9	53.9	48.9	46.4	50.3	54.2	62.6	69.1	70.5
		MCWB	60.2	60.0	57.5	52.9	48.5	45.0	43.3	45.5	47.0	51.5	55.8	57.1
	5%	DB	70.2	68.6	64.5	57.3	51.0	46.5	44.4	47.1	50.9	58.1	63.6	65.4
		MCWB	58.1	58.4	55.8	50.8	47.5	43.7	41.9	43.6	45.6	49.9	53.4	55.4
	10%	DB	65.6	64.4	60.5	54.2	48.8	44.7	42.9	44.6	47.8	53.4	59.0	60.9
		MCWB	56.1	56.4	54.5	49.8	46.3	42.7	41.1	42.0	44.1	47.9	51.4	53.8
	0.4%	WB	64.2	64.3	61.7	56.2	52.0	49.8	46.0	48.6	51.4	56.3	60.2	61.5
		MCDB	75.2	75.1	71.3	61.6	54.9	50.6	47.9	52.8	57.3	63.8	70.5	71.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	61.3	61.7	58.9	54.1	49.9	46.3	43.9	46.1	48.4	53.1	57.2	58.8
		MCDB	71.8	70.5	66.5	58.8	52.2	47.9	45.6	49.3	52.9	60.4	66.5	67.9
	5%	WB	59.1	59.4	56.9	52.1	48.2	44.3	42.6	44.2	46.2	50.7	54.5	56.4
		MCDB	69.3	66.9	62.9	56.4	50.5	45.6	43.9	46.9	49.8	57.1	62.3	64.2
	10%	WB	56.7	57.2	54.9	50.0	46.5	43.1	41.5	42.4	44.2	48.2	51.9	54.1
		MCDB	64.5	63.5	60.0	53.5	48.5	44.2	42.6	44.3	47.2	53.4	58.6	60.4
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	15.9	13.1	10.1	7.6	6.6	6.6	7.8	10.3	13.0	15.0	15.3
		MCDBR	25.5	22.8	19.9	15.2	10.7	9.0	8.8	12.2	15.9	21.1	23.7	23.6
		MCWBR	12.9	12.2	11.7	9.7	7.8	6.9	6.9	8.5	10.8	12.5	13.6	13.5
	5% WB	MCDBR	23.3	20.5	18.1	14.0	9.6	7.4	8.0	11.5	14.4	19.3	21.9	21.6
		MCWBR	12.7	12.0	11.5	9.7	7.6	6.4	6.7	8.4	10.5	12.4	13.5	13.3
(40) Clear-Sky Solar Irradiance	taub	0.316	0.304	0.303	0.290	0.282	0.272	0.275	0.287	0.308	0.311	0.308	0.313	(40)
	taud	2.566	2.619	2.613	2.629	2.630	2.638	2.637	2.613	2.533	2.536	2.570	2.564	(41)
	Ebn at Noon	320	316	302	284	264	255	264	281	295	310	321	323	(42)
	Edh at Noon	33	30	27	23	19	17	18	23	29	32	32	33	(43)
(44) All-Sky Solar Radiation	RadAvg	1976	1697	1232	826	497	397	456	683	1034	1483	1807	1975	(44)
	RadStd	141	125	74	48	37	36	28	49	52	82	137	120	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		+0.59	N/A	N/A	+1.21	N/A	N/A	N/A	-180	+191	+25

Nomenclature: See separate page